b. **Beacon by Leadrat**

Complete documentation

Live demo · how it works · technology stack · detailed cost analysis — prices as of 16 October 2026

Beacon is an AI product guide for the Leadrat website. A visitor clicks **Start exploring**, sees the real Leadrat CRM live on the left of a popup and talks to Beacon on the right, by typing or by voice. Beacon answers from the company's product handbook, opens and highlights the matching CRM screen while it explains, speaks the answer, asks follow-up questions about the visitor's business, and saves the visitor as a lead in the CRM tenant (today: **surya**).

At a glance

AI MODEL

DeepSeek V4.1 Flash

3 calls per question: understand, answer from the handbook, check the answer

VECTOR DATABASE

PostgreSQL + pgvector

384-dimension vectors, multilingual (incl. Hinglish), hybrid with keyword search

VOICE

Soniox (or free Edge voice)

Speech-to-text stt-rt-v5 · text-to-speech tts-rt-v2

LEADS

Saved to tenant “surya”

Leadrat API; any tenant can be configured

Cost at a glance (1,000 users / month, 20 questions each)	Per month	Per year
A · Free Microsoft Edge voice + AWS hosting	**$183.64 (₹16,528)**	$2,203.70 (₹198,333)
B · Soniox voice + AWS hosting	**$289.80 (₹26,082)**	$3,477.54 (₹312,979)
C · Soniox voice + visitors speak their questions	**$317.13 (₹28,542)**	$3,805.54 (₹342,499)
One product question, averaged over a visit (Edge voice / Soniox voice)	$0.0019 (₹0.17) / $0.0072 (₹0.65)	

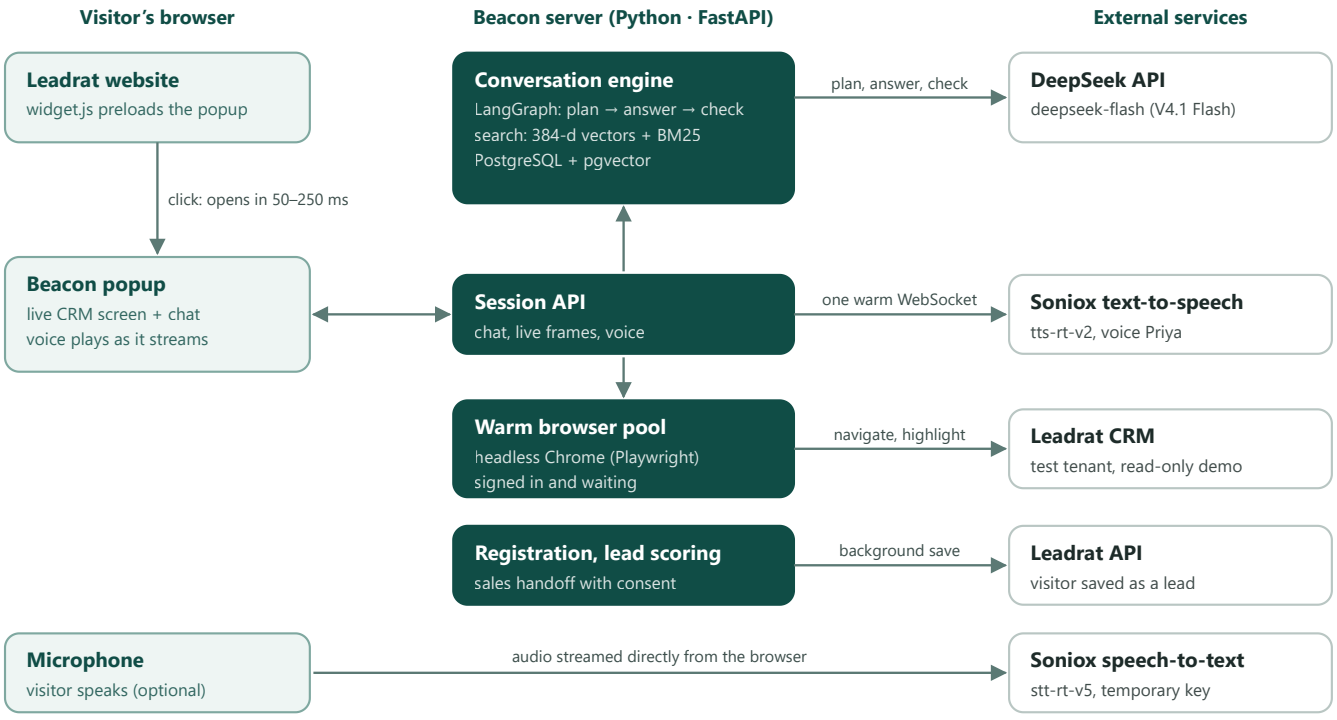
Prices exclude GST. $1 = ₹90 (planning rate). DeepSeek at peak prices (worst case). Full details in section 8.

1. What the demo video shows

The video `Beacon_Demo_Video.mp4` (7 min 27 s) is a real screen recording of the product with Beacon's own voice, plus a short voice-over and animated explanations.

Time	What happens	What it proves
0:00	Intro: what Beacon is	—
0:14	Visitor fills name, work email and phone; Beacon greets them by name	Registration in the widget, phone validated with country code
0:42	The Surya CRM: the new lead appears with name, phone, email and Source = Website	Details are saved to a specific tenant (surya); any tenant can be configured
0:59	Typed question: “tell me about Sales Copilot”	Beacon opens the Sales Copilot page live and answers only from the handbook
1:46	Spoken question about bulk operations (microphone)	Soniox speech-to-text; Beacon selects leads and shows bulk actions without running them
2:47	Question about Leadrat AI	Opens the Leadrat AI page, explains tenant permissions
3:29	Beacon asks its follow-up questions; the visitor answers	Business type, team size, monthly leads, tools, problem, decision, timeline, contact consent → lead score
4:57	“How do I add a lead?”	Opens the Add Lead form; saves only after the visitor confirms
5:41	Animated explanation of the architecture, tech stack, vector database and costs	Sections 2–8 of this document

2. How it works



1. The widget sends the visitor's message. If Beacon is still speaking, the new question interrupts it.
2. Local search (≈5 ms): hybrid search over the handbook and the CRM screen map (2,054 screens from the CRM code + 38 prepared walkthroughs).

3. **DeepSeek "plan"**: decides what the visitor means (product question, business answer, small talk, refusal, data-change request) and which screen to show.
4. The hidden Chrome browser opens and highlights that screen **while** the answer is written.
5. **DeepSeek "answer"**: writes the reply only from selected handbook passages and screen facts.
6. **DeepSeek "check"**: verifies every claim against those passages; meanwhile the voice is already being generated.
7. The widget shows the text and plays the voice as it streams; the live screen shows the walkthrough.

Safety rules in the code: the demo browser can only navigate and highlight; the only write is Add Lead, saved after an explicit confirmation; bulk actions and Delete are pointed at, never clicked; a command to change CRM data is refused; contact details go to sales only with consent.

3. Technology stack

Layer	Technology	Used for
Backend	Python 3.11, FastAPI 0.141, Uvicorn 0.53	Session API, widget files
Conversation engine	LangGraph 1.2	Turn pipeline: context → plan → grounded answer → check → save
Language model	DeepSeek API, `deepseek-flash` (V4.1 Flash)	Understanding, answers, source checks, lead facts (JSON mode)
Live CRM	Playwright 1.58 + Google Chrome (headless)	Signs in, navigates, highlights; frames sent as JPEG
Embeddings	paraphrase-multilingual-MiniLM-L12-v2, INT8 ONNX on ONNX Runtime	384-dimension vectors, 50+ languages incl. Hinglish
Vector database	PostgreSQL + pgvector (deployment); SQLite + NumPy (local today)	Handbook passages, conversation state
Speech-to-text	Soniox real-time `stt-rt-v5`	Visitor voice → text (English, Hindi, Hinglish)
Noise cleanup	RNNoise (web-noise-suppressor 0.4.1)	Removes background noise before Soniox
Text-to-speech	Soniox real-time `tts-rt-v2`, voice Priya	Beacon's spoken replies (streamed MP3)
Backup / free voice	Microsoft Edge read-aloud (`edge-tts` 7.3.1)	Free voice today; backup if Soniox fails
Widget	HTML, CSS, JavaScript, intl-tel-input	Popup UI, phone field, streaming audio
Registration	Leadrat API (tenant **surya**), phonenumbers	Visitor saved as a CRM lead in the background

4. Knowledge search and the vector database

Item	Value
Embedding model	sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2, INT8-quantised ONNX, runs on the CPU (no GPU, no per-call cost)
Vector dimension	**384** (float32), normalised, cosine similarity
Handbook	FLOE Leadrat Pre-Sales Product Knowledge Handbook v0.2 (29 pages) → 139 passages, 13 modules
CRM screens	2,054 screens/sections + 38 walkthroughs = 2,092 entries × 384 dimensions
Database (deployment)	PostgreSQL with **pgvector**: column `vector(384)`, cosine distance operator `<=>`; exact search
Database (local today)	SQLite file + in-memory NumPy search
Search method	Hybrid: vector similarity + BM25 keywords, merged with reciprocal rank fusion (k = 60); Hinglish is romanised first

Item	Value
Speed	≈5 ms of CPU per question

The model never sees the whole handbook: each question gets the best few passages, the model must cite them, and a second call checks the answer.

5. Voice

- **Visitor → Beacon:** Soniox `stt-rt-v5` streamed from the browser with a single-use 60-second key (the real key never reaches the browser). RNNoise cleans the microphone first.
- **Beacon → visitor:** Soniox `tts-rt-v2` (voice Priya) over one warm WebSocket; up to five sentences generated in parallel; audio plays while it is still being generated.
- **Fixed sentences** (welcome, follow-up and business questions) are generated once and cached, so they cost nothing per visitor.
- **Microsoft Edge voice** is free but unofficial (no contract or guarantee; an older version was blocked with HTTP 403 in October 2026). Soniox is the licensed, paid option.

6. Measured speed and resources

What	Measured
Click “Start exploring” → live CRM visible	55–245 ms
Product question → answer text	3.3–4.7 s (three DeepSeek calls)
Answer text → voice starts (Soniox)	45–77 ms
Memory per live demo	1.45 GB at start, 1.61 GB after 8 questions
CPU per live demo	≈0.3 core while the visitor reads, up to 0.94 core while answering
Data sent to the visitor	≈0.9 MB per product question

7. Leads, tenant and qualification

- The registration form (name, work email, phone with country code) is validated and saved through the Leadrat API to the configured tenant — in the demo, **surya** (Surya QA). The tenant, API address and credentials are settings, so any company tenant can be used.
- The lead is created with Source = Website; a duplicate check by phone number runs first; the demo continues without waiting for the save.
- Beacon then asks business questions once each — business type, team size, monthly leads, current tool, biggest problem, decision role, timeline, sales contact — and scores the lead. Sales contact happens only with the visitor's consent.

8. Detailed cost analysis

Beacon has three kinds of cost: **AI** (DeepSeek, per question), **voice** (free Microsoft Edge voice, or Soniox per spoken character; optional Soniox speech-to-text when visitors speak), and **deployment** (an AWS server in Mumbai, a fixed monthly bill sized for how many demos run at the same time). Three options are compared everywhere:

Option	Voice the visitor hears	Visitor input	Licensing
A	Microsoft Edge voice — free	Typing	Unofficial, no guarantee
B	Soniox voice (Priya)	Typing	Licensed, paid per use
C	Soniox voice (Priya)	Speaking (microphone)	Licensed, paid per use

8.1 Prices used

Service	Price	Source
DeepSeek `deepseek-flash`	Per 1M tokens (peak): cached input $0.006, uncached input $0.30, output $1.20. Off-peak is half.	api-docs.deepseek.com/quick_start/pricing
Soniox text-to-speech `tts-rt-v2`	$4 / 1M text tokens + $21.50 / 1M audio tokens ≈ **$0.70 per hour of speech** ≈ $10.58 per 1M spoken characters	soniox.com/pricing
Soniox speech-to-text `stt-rt-v5`	**$0.12 per hour** of audio	soniox.com/pricing
Microsoft Edge voice	**Free**	—
AWS EC2 Mumbai (on demand, Linux)	c6a.xlarge $0.0935/h · c6a.2xlarge $0.187/h · m6a.2xlarge $0.2222/h · c6a.4xlarge $0.374/h	aws.amazon.com/ec2/pricing
AWS disk, backups, IP	EBS gp3 $0.0912/GB-month · snapshots $0.05/GB-month · IPv4 $0.005/h	aws.amazon.com/ebs/pricing, /vpc/pricing
AWS data to the internet	First 100 GB a month free	aws.amazon.com/ec2/pricing

DeepSeek peak hours: 01:00–04:00 and 06:00–10:00 UTC Mon–Fri (06:30–09:30 and 11:30–15:30 IST). Repeated prompt parts are cached, which is why a question that sends ≈6,000 prompt tokens costs so little.

8.2 Cost of one product question

Part of one product question	Cost (peak prices)	When it applies
DeepSeek · understand & plan (≈1.25 calls)	$0.00062 (₹0.06)	Every question
DeepSeek · grounded answer from the handbook	$0.00077 (₹0.07)	Every question
DeepSeek · source check of the answer	$0.00023 (₹0.02)	Every question
AI subtotal	**$0.00162 (₹0.15)**	
Spoken reply · Microsoft Edge voice (479 characters)	$0.00000 (₹0.00)	Option A (today)
Spoken reply · Soniox voice (479 characters)	$0.00507 (₹0.46)	Option B
Visitor speaks the question · Soniox speech-to-text (36 s of audio)	$0.00120 (₹0.11)	Only if the visitor uses the mic
Total · Option A (Edge voice, typed)	**$0.00162 (₹0.15)**	
Total · Option B (Soniox voice, typed)	**$0.00669 (₹0.60)**	
Total · Option C (Soniox voice + spoken question)	**$0.00789 (₹0.71)**	

Answers to Beacon's business questions are cheaper: $0.00080 (₹0.07) AI and $0.00047 (₹0.04) Soniox voice each. The welcome and Beacon's fixed questions are cached and cost nothing per visitor. Averaged over a full visit (each question plus the business answers and

registration around it), one question costs $0.0019 (₹0.17) with the Edge voice and $0.0072 (₹0.65) with the Soniox voice — the ₹0.65 shown in the video.

8.3 Cost of one visitor journey (without hosting)

One visitor registers, asks N product questions and, if they ask any, answers Beacon's 8 business questions once.

Product questions	Time in demo	DeepSeek (AI)	Soniox voice	Speech-to-text	A · Edge voice	B · Soniox voice	C · Soniox + speech
0	2 min	$0.0000 (₹0.00)	$0.0011 (₹0.10)	$0.0000 (₹0.00)	**$0.0000 (₹0.00)**	**$0.0011 (₹0.10)**	**$0.0011 (₹0.10)**
5	8 min	$0.0145 (₹1.31)	$0.0301 (₹2.71)	$0.0093 (₹0.84)	**$0.0145 (₹1.31)**	**$0.0447 (₹4.02)**	**$0.0540 (₹4.86)**
10	12 min	$0.0226 (₹2.04)	$0.0555 (₹4.99)	$0.0153 (₹1.38)	**$0.0226 (₹2.04)**	**$0.0781 (₹7.03)**	**$0.0934 (₹8.41)**
20	19 min	$0.0388 (₹3.50)	$0.1062 (₹9.55)	$0.0273 (₹2.46)	**$0.0388 (₹3.50)**	**$0.1450 (₹13.05)**	**$0.1723 (₹15.51)**
30	26 min	$0.0550 (₹4.95)	$0.1568 (₹14.12)	$0.0394 (₹3.54)	**$0.0550 (₹4.95)**	**$0.2119 (₹19.07)**	**$0.2512 (₹22.61)**
50	42 min	$0.0875 (₹7.87)	$0.2582 (₹23.24)	$0.0634 (₹5.71)	**$0.0875 (₹7.87)**	**$0.3456 (₹31.11)**	**$0.4091 (₹36.82)**

8.4 Deployment cost (hosting)

Each live demo needs its own signed-in Chrome browser: measured ≈1.6 GB of memory and up to ≈0.9 CPU core while answering. The server is sized for the busiest hour (twice the average traffic, working days, 10 hours a day), so 95% of busiest-hour moments are served at once, plus one spare demo kept ready so "Start exploring" stays instant. The cheapest AWS Mumbai server that fits is chosen.

Questions per user	Demo hours / month	Demos at once*	AWS server (Mumbai)	Server	Disk, backups, IP	Hosting / month	Data out
0	33	2	c6a.xlarge 4 vCPU · 8 GB · up to 3 demos	$68.25 (₹6,143)	$8.30 (₹747)	**$76.55 (₹6,890)**	0 GB
10	192	5	c6a.2xlarge 8 vCPU · 16 GB · up to 8 demos	$136.51 (₹12,286)	$8.30 (₹747)	**$144.81 (₹13,033)**	9 GB
20	317	7	c6a.2xlarge 8 vCPU · 16 GB · up to 8 demos	$136.51 (₹12,286)	$8.30 (₹747)	**$144.81 (₹13,033)**	18 GB
30	442	9	m6a.2xlarge 8 vCPU · 32 GB · up to 10 demos	$162.21 (₹14,599)	$8.30 (₹747)	**$170.50 (₹15,345)**	27 GB
50	692	12	c6a.4xlarge 16 vCPU · 32 GB · up to 17 demos	$273.02 (₹24,572)	$8.30 (₹747)	**$281.32 (₹25,319)**	45 GB

* Demos running at the same time in the busiest hour, plus one ready. Visitor pace assumed: 2 min to register, 45 s per product question, 15 s per business answer.

Fixed part of every deployment	Size	Price	Per month
System disk (EBS gp3)	40 GB	$0.0912 / GB-month	$3.65 (₹328)
Daily backups (EBS snapshots)	≈20 GB	$0.05 / GB-month	$1.00 (₹90)
Public IPv4 address	1	$0.005 / hour	$3.65 (₹328)
PostgreSQL + pgvector, web proxy, HTTPS (Let's Encrypt)	same server	free	$0.00
Data sent to visitors	up to 100 GB	first 100 GB free	$0.00
Total fixed extras			**$8.30 (₹747)**

Cheaper hosting alternatives (not in the totals): DigitalOcean Bangalore — 8 vCPU / 16 GB Basic droplet $96/month (shared CPU, disk and 6 TB data included), CPU-Optimized $168/month; an AWS 1-year Savings Plan also lowers the

server price. **One-time engineering (no bill):** today one server process runs one demo; production needs a small router that gives each visitor a free worker, and a "please wait" message when all are busy.

8.5 Monthly total for 1,000 users — all options, deployment included

Questions per user	Hosting (AWS)	DeepSeek (AI)	Soniox voice	Speech-to-text	A · Edge voice (free) + hosting	B · Soniox voice + hosting	C · Soniox voice + speech input
0	$76.55 (₹6,890)	$0.00 (₹0)	$1.07 (₹96)	$0.00 (₹0)	**$76.55 (₹6,890)**	**$77.62 (₹6,986)**	**$77.62 (₹6,986)**
10	$144.81 (₹13,033)	$22.63 (₹2,036)	$55.47 (₹4,993)	$15.31 (₹1,378)	**$167.44 (₹15,069)**	**$222.91 (₹20,062)**	**$238.22 (₹21,439)**
20	$144.81 (₹13,033)	$38.83 (₹3,495)	$106.15 (₹9,554)	$27.33 (₹2,460)	**$183.64 (₹16,528)**	**$289.80 (₹26,082)**	**$317.13 (₹28,542)**
30	$170.50 (₹15,345)	$55.04 (₹4,954)	$156.83 (₹14,115)	$39.36 (₹3,542)	**$225.54 (₹20,299)**	**$382.38 (₹34,414)**	**$421.74 (₹37,956)**
50	$281.32 (₹25,319)	$87.45 (₹7,871)	$258.19 (₹23,238)	$63.41 (₹5,707)	**$368.77 (₹33,189)**	**$626.97 (₹56,427)**	**$690.38 (₹62,134)**

Where the money goes (1,000 users × 20 questions)	Option A · Edge voice	Share	Option B · Soniox voice	Share
Hosting (AWS Mumbai)	$144.81 (₹13,033)	79%	$144.81 (₹13,033)	50%
DeepSeek (AI answers)	$38.83 (₹3,495)	21%	$38.83 (₹3,495)	13%
Voice (text-to-speech)	$0.00 (free)	0%	$106.15 (₹9,554)	37%
Total per month	**$183.64 (₹16,528)**	100%	**$289.80 (₹26,082)**	100%

Reading the numbers: hosting is the largest fixed cost (≈$77/month even with no questions). With the free Edge voice, AI is the only usage cost. With Soniox, the voice becomes the largest usage cost — about $0.0051 (₹0.46) per spoken answer versus $0.0016 (₹0.15) for all AI calls behind it. Moving from Edge to Soniox adds $106.15 (₹9,554) a month at 20 questions per user.

8.6 Scaling with the number of users (20 questions each)

Users / month	Server	Hosting	DeepSeek	Soniox voice	A · Edge voice	B · Soniox voice	Per user (A / B)
250	c6a.xlarge	$76.55 (₹6,890)	$9.71 (₹874)	$26.54 (₹2,388)	**$86.26 (₹7,764)**	**$112.80 (₹10,152)**	₹31.1 / ₹40.6
500	c6a.2xlarge	$144.81 (₹13,033)	$19.42 (₹1,748)	$53.08 (₹4,777)	**$164.22 (₹14,780)**	**$217.30 (₹19,557)**	₹29.6 / ₹39.1
1,000	c6a.2xlarge	$144.81 (₹13,033)	$38.83 (₹3,495)	$106.15 (₹9,554)	**$183.64 (₹16,528)**	**$289.80 (₹26,082)**	₹16.5 / ₹26.1
2,500	c6a.4xlarge	$281.32 (₹25,319)	$97.08 (₹8,738)	$265.38 (₹23,885)	**$378.40 (₹34,056)**	**$643.79 (₹57,941)**	₹13.6 / ₹23.2
5,000	2 × c6a.4xlarge	$562.64 (₹50,637)	$194.17 (₹17,475)	$530.77 (₹47,769)	**$756.80 (₹68,112)**	**$1,287.57 (₹115,881)**	₹13.6 / ₹23.2

More users need more demos at the same time, so the server grows in steps. Data sent to visitors stays inside AWS's free 100 GB up to 5,000 users × 20 questions (≈90 GB).

8.7 Annual cost for 1,000 users a month

Questions per user	A · Edge voice / year	B · Soniox voice / year	C · Soniox + speech / year	Extra for Soniox voice / year
0	$918.64 (₹82,677)	$931.46 (₹83,831)	$931.46 (₹83,831)	$12.82 (₹1,154)
20	$2,203.70 (₹198,333)	$3,477.54 (₹312,979)	$3,805.54 (₹342,499)	$1,273.84 (₹114,646)
30	$2,706.53 (₹243,587)	$4,588.53 (₹412,968)	$5,060.85 (₹455,476)	$1,882.01 (₹169,381)
50	$4,425.24 (₹398,272)	$7,523.58 (₹677,122)	$8,284.52 (₹745,607)	$3,098.34 (₹278,850)

8.8 Saving by running outside DeepSeek peak hours

Questions per user	DeepSeek at peak	DeepSeek off-peak	Saving / month	A · Edge total off-peak	B · Soniox total off-peak
20	$38.83 (₹3,495)	$19.42 (₹1,748)	$19.42 (₹1,748)	$164.22 (₹14,780)	$270.38 (₹24,334)
30	$55.04 (₹4,954)	$27.52 (₹2,477)	$27.52 (₹2,477)	$198.02 (₹17,822)	$354.86 (₹31,937)
50	$87.45 (₹7,871)	$43.73 (₹3,935)	$43.73 (₹3,935)	$325.04 (₹29,254)	$583.24 (₹52,492)

8.9 Edge voice vs Soniox voice

	A · Microsoft Edge voice	B/C · Soniox voice
Price	Free	≈$0.0051 (₹0.46) per spoken answer; $106.15 (₹9,554) / month at 1,000 users × 20 questions
Licence and reliability	Unofficial use of a browser read-aloud service; no contract; can be blocked (HTTP 403 seen in Oct 2026)	Commercial API with an account and pricing
Speed	First audio 530–600 ms	First audio 480–550 ms; 45–77 ms after the text appears (generated during the source check)
Voice	en-IN Neerja	Priya (Indian English), consistent with Soniox speech-to-text
Recommendation	Fine for internal demos and testing	Recommended for the public website

8.10 Not included in the totals

- Taxes (e.g. 18% GST), discounts and credits.
- Leadrat CRM licence — the demo uses an existing test tenant.
- A second server for redundancy (roughly doubles the server price).
- Engineering time for the multi-demo router and deployment.
- Data transfer above 100 GB a month (only beyond ≈5,000 users × 20 questions).

9. Assumptions and sources

- Measured on 16 October 2026: a 16-turn session (8 product questions + 8 business answers, 36 DeepSeek calls, 4,185 new spoken characters) and an 8-question session for memory, CPU and bandwidth.
- Spoken characters: 479 per product answer, 44 per business answer, 101 for the registration reply; Soniox Priya at speed 1.1 speaks 19.1 characters per second.
- Visitors arrive on working days, 10 hours a day; the busiest hour has twice the average traffic; 95% of busiest-hour moments served at once plus one ready demo.
- Exchange rate $1 = ₹90; prices exclude taxes; DeepSeek at peak prices unless stated.
- All tables are produced by `eval/cost_model.py`; re-run `python docs/build_complete_pdf.py` after any price change.

Sources: api-docs.deepseek.com/quick_start/pricing · soniox.com/pricing · soniox.com/docs · aws.amazon.com/ec2/pricing/on-demand · aws.amazon.com/ebs/pricing · aws.amazon.com/vpc/pricing · digitalocean.com/pricing/droplets